

Exchange rate response to news

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Keywords:

JEL Codes:

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1 Introduction

Coleman and Karagedikli (2012)

2 Literature review

3 Data and methodology

4 Results

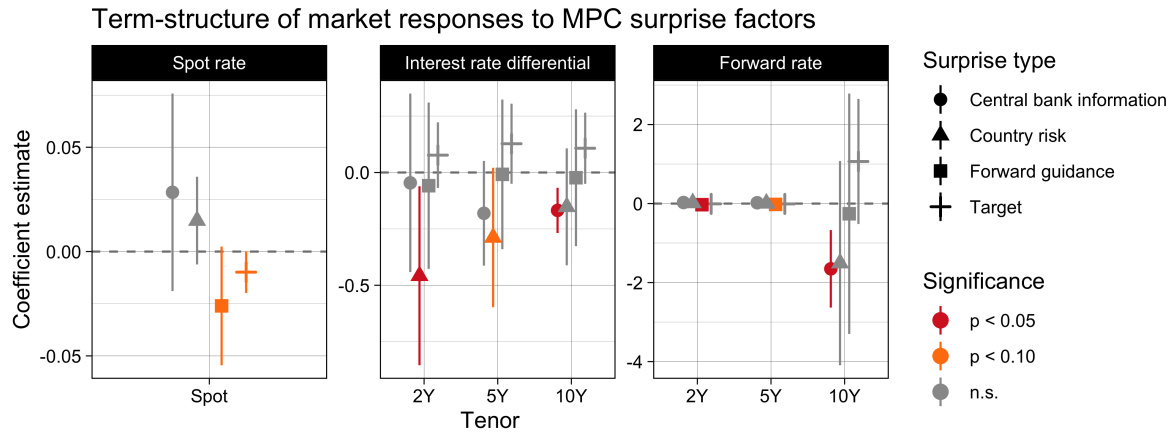


Figure 1: Response to Target

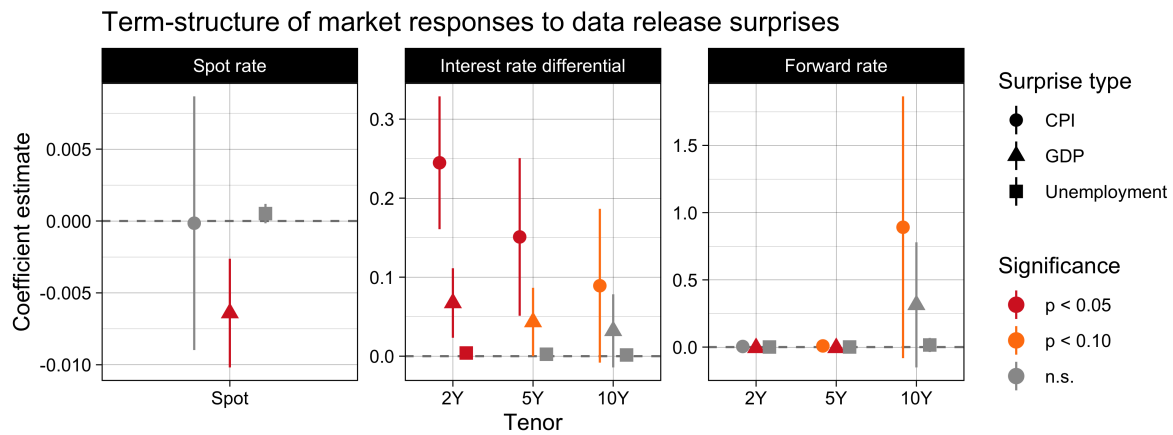


Figure 2: Data release response

Residual-on-residual: spot vs OIS

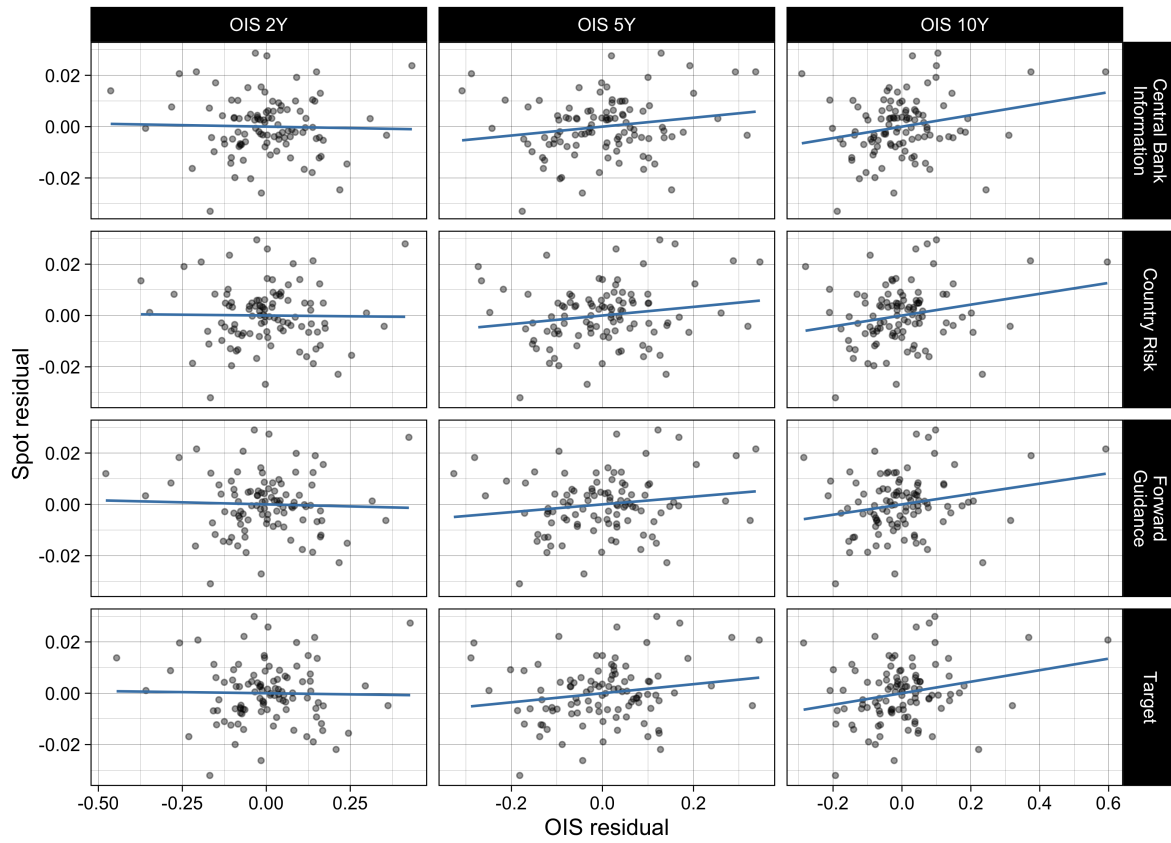


Figure 3: Spot residuals versus interest rate residual

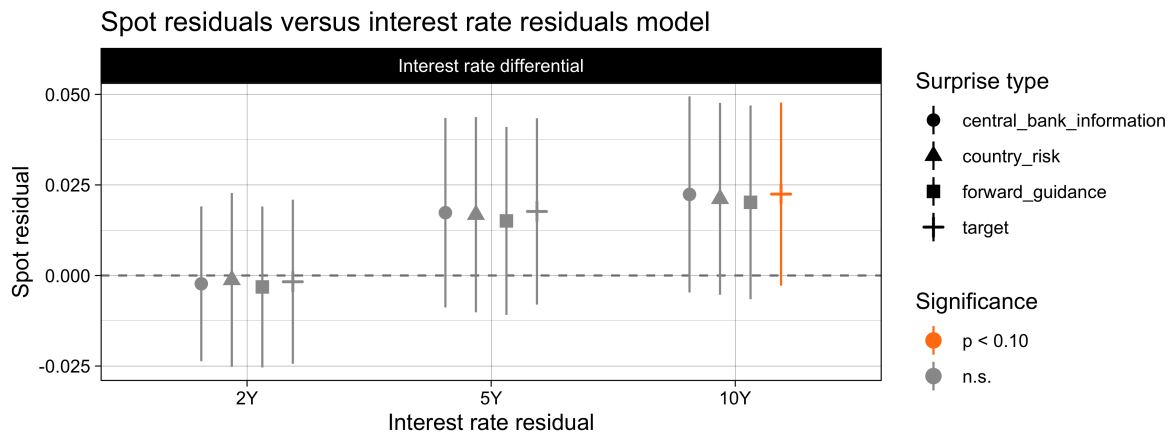


Figure 4: Spot residuals versus interest rate residuals model

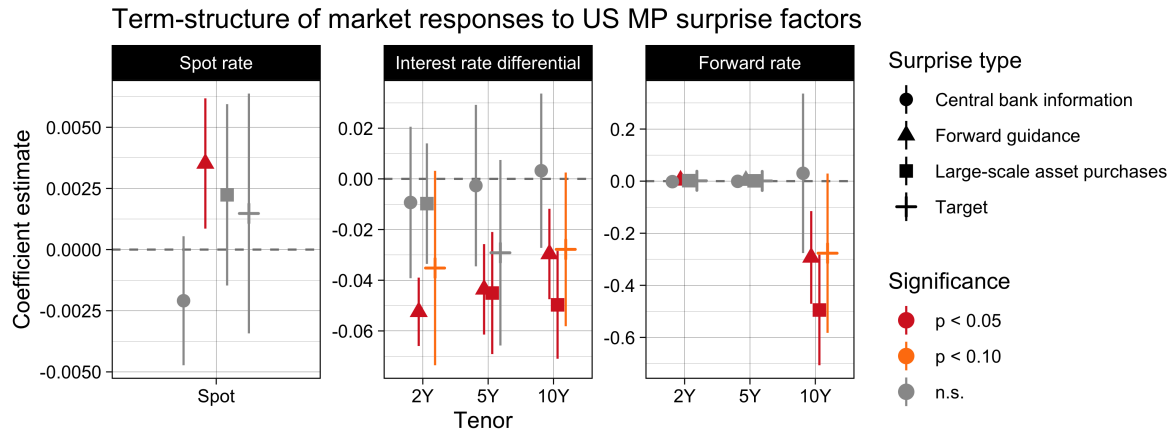


Figure 5: US MPS response

5 Conclusion

References

Coleman, A. and Karagedikli, Ö. (2012). The relative size of exchange rate and interest rate responses to news: An empirical investigation. *The North American Journal of Economics and Finance*, 23(1):1–19.

A Appendix

A.1 Data

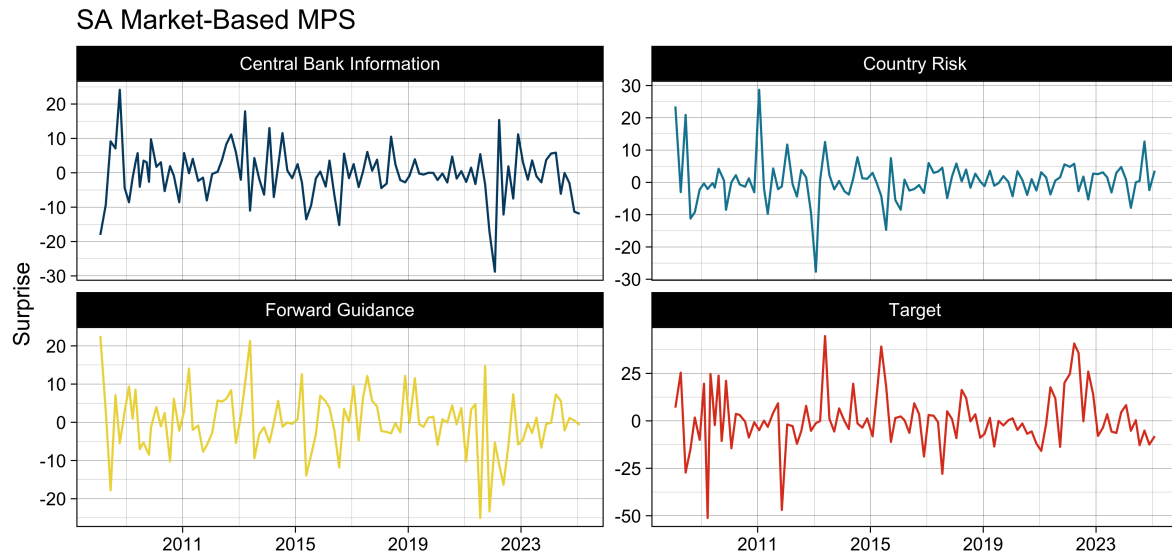


Figure A1: SA monetary policy surprises

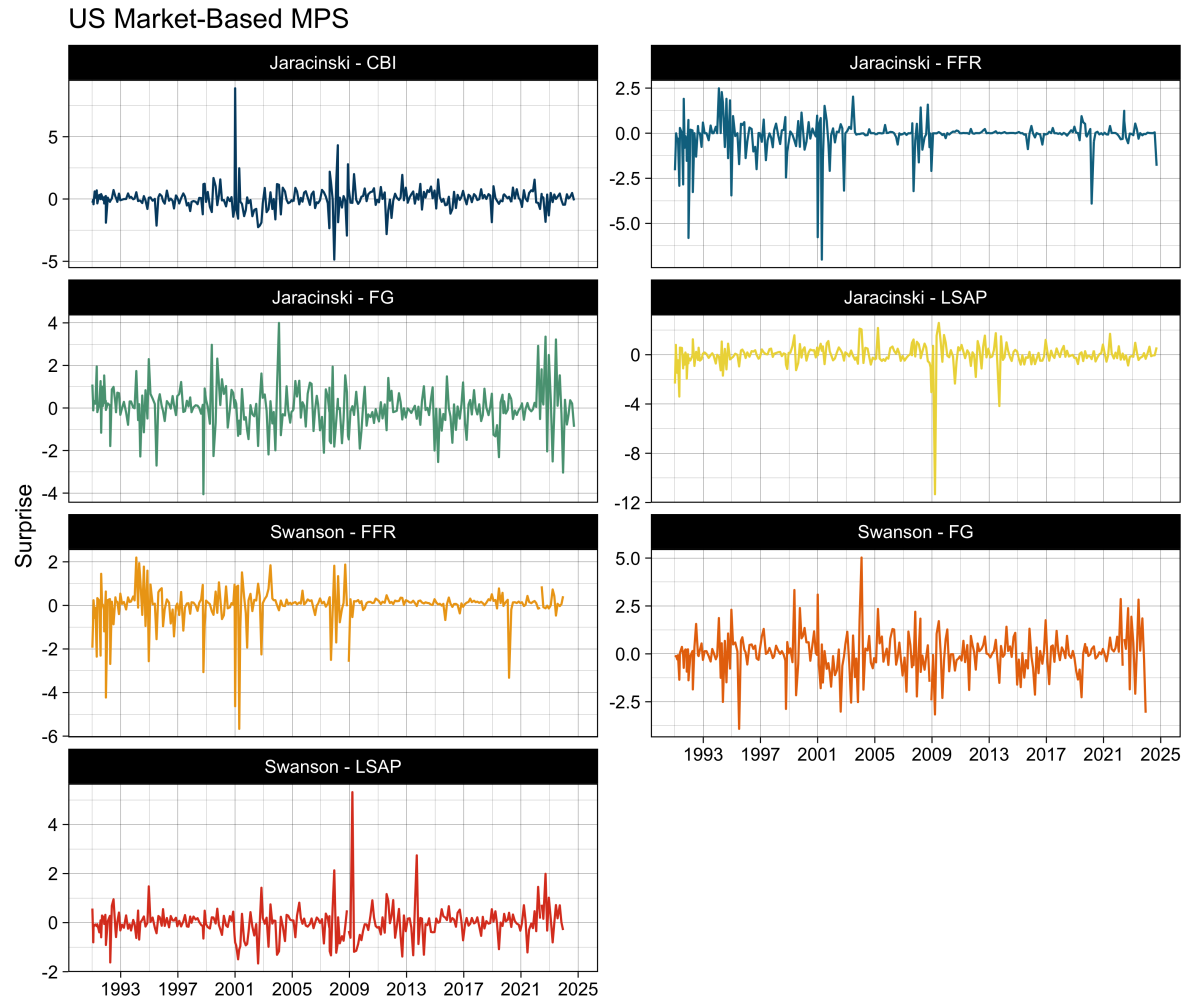


Figure A2: US monetary policy surprises

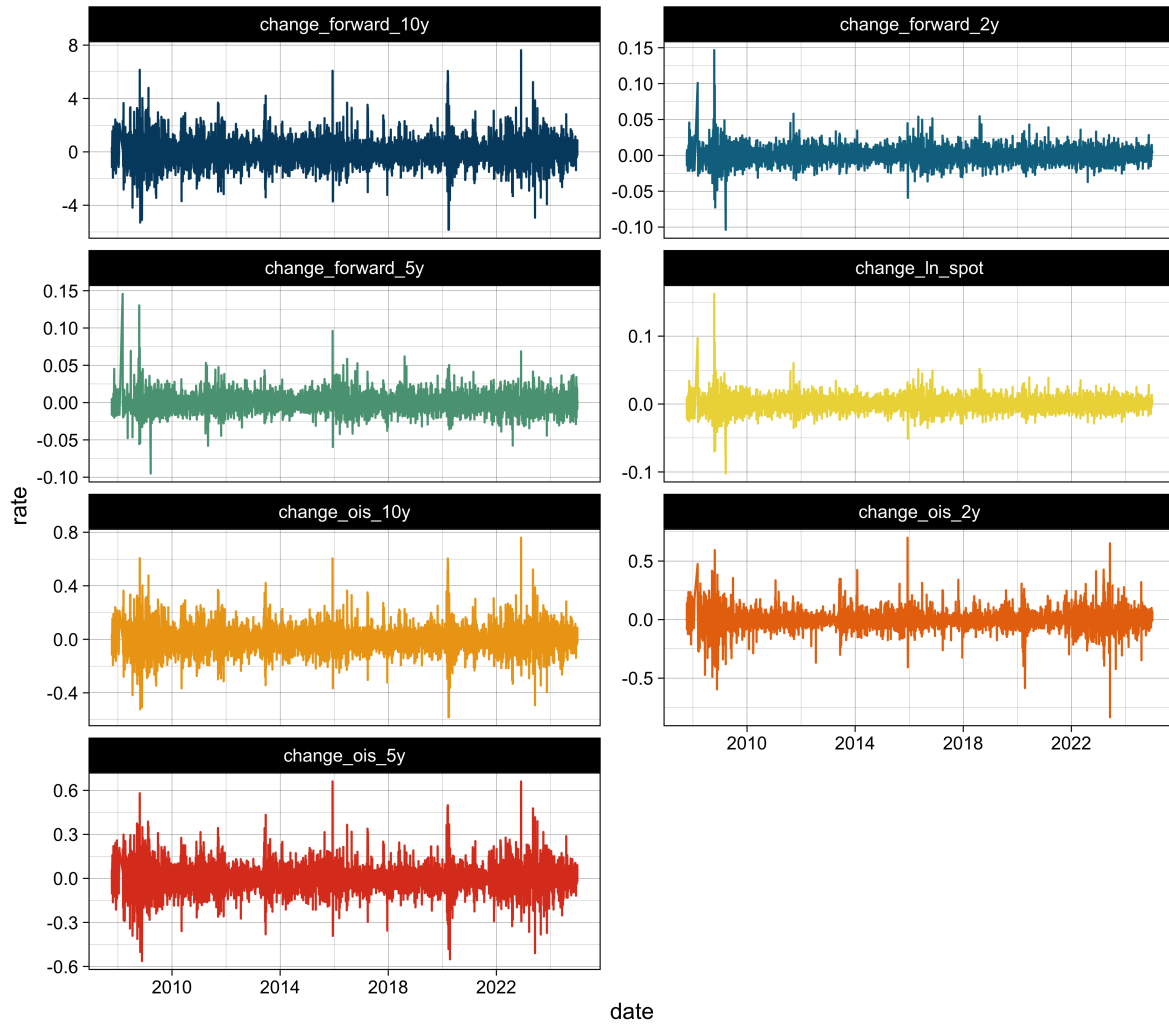


Figure A3: Key variables

Table A1: Descriptive statistics

series	Mean	Median	SD	Min	Max	Observations
change_forward_10y	0.0029	-0.0155	1.0217	-5.8376	7.6202	4,137
change_forward_2y	0.0002	-0.0002	0.0117	-0.1038	0.1466	4,137
change_forward_5y	0.0002	-0.0004	0.0132	-0.0951	0.1457	4,137
change_ln_spot	0.0002	0.0000	0.0114	-0.1023	0.1621	4,137
change_ois_10y	0.0003	-0.0017	0.1018	-0.5827	0.7593	4,137
change_ois_2y	-0.0005	-0.0028	0.0862	-0.8343	0.6997	4,137
change_ois_5y	-0.0002	-0.0021	0.0990	-0.5631	0.6607	4,137

B Detailed results

Table A2: SA monetary policy surprises models

surprise type	dependant variables	estimate	std.error	statistic	p.value	r.squared	nobs
target models	change_ln_spot	-0.010	0.005	-1.931	0.056	0.017	101
	change_ois_2y	0.077	0.074	1.037	0.302	0.007	101
	change_ois_5y	0.127	0.091	1.407	0.163	0.027	101
	change_ois_10y	0.108	0.081	1.335	0.185	0.018	101
	change_forward_2y	-0.007	0.005	-1.387	0.169	0.009	101
	change_forward_5y	-0.009	0.006	-1.556	0.123	0.010	101
	change_forward_10y	1.065	0.807	1.320	0.190	0.017	101
forward guidance models	change_ln_spot	-0.026	0.015	-1.793	0.076	0.030	101
	change_ois_2y	-0.059	0.188	-0.311	0.756	0.001	101
	change_ois_5y	-0.008	0.169	-0.049	0.961	0.000	101
	change_ois_10y	-0.023	0.155	-0.150	0.881	0.000	101
	change_forward_2y	-0.028	0.013	-2.109	0.037	0.036	101
	change_forward_5y	-0.021	0.012	-1.747	0.084	0.016	101
	change_forward_10y	-0.258	1.552	-0.166	0.868	0.000	101
central bank information models	change_ln_spot	0.028	0.024	1.176	0.242	0.033	101
	change_ois_2y	-0.046	0.202	-0.227	0.821	0.001	101
	change_ois_5y	-0.181	0.118	-1.527	0.130	0.012	101
	change_ois_10y	-0.168	0.051	-3.281	0.001	0.010	101
	change_forward_2y	0.027	0.021	1.258	0.211	0.029	101
	change_forward_5y	0.021	0.022	0.973	0.333	0.014	101
	change_forward_10y	-1.652	0.500	-3.303	0.001	0.010	101
country risk models	change_ln_spot	0.015	0.011	1.386	0.169	0.007	101
	change_ois_2y	-0.458	0.202	-2.261	0.026	0.047	101
	change_ois_5y	-0.288	0.158	-1.828	0.071	0.024	101
	change_ois_10y	-0.152	0.132	-1.151	0.253	0.006	101
	change_forward_2y	0.015	0.013	1.204	0.232	0.007	101
	change_forward_5y	0.005	0.018	0.311	0.757	0.001	101
	change_forward_10y	-1.507	1.319	-1.143	0.256	0.006	101

Table A3: SA residuals models

surprise type	predictor_model	term	estimate	std.error	statistic	p.value	r.squared	nobs
central bank information models								
	change_ois_2y	resid_predictor	-0.002	0.011	-0.210	0.834	0.001	101
	change_ois_5y	resid_predictor	0.017	0.013	1.298	0.197	0.033	101
	change_ois_10y	resid_predictor	0.022	0.014	1.620	0.108	0.058	101
country risk models								
	change_ois_2y	resid_predictor	-0.001	0.012	-0.100	0.921	0.000	101
	change_ois_5y	resid_predictor	0.017	0.014	1.221	0.225	0.029	101
	change_ois_10y	resid_predictor	0.021	0.014	1.567	0.120	0.050	101
forward guidance models								
	change_ois_2y	resid_predictor	-0.003	0.011	-0.279	0.781	0.001	101
	change_ois_5y	resid_predictor	0.015	0.013	1.138	0.258	0.025	101
	change_ois_10y	resid_predictor	0.020	0.014	1.480	0.142	0.047	101
target models								
	change_ois_2y	resid_predictor	-0.002	0.012	-0.150	0.881	0.000	101
	change_ois_5y	resid_predictor	0.018	0.013	1.347	0.181	0.033	101
	change_ois_10y	resid_predictor	0.022	0.013	1.742	0.085	0.057	101

Table A4: SA data release surprises models

surprise type	dependant variables	estimate	std.error	statistic	p.value	r.squared	nobs
cpi models	change_ln_spot	-0.000	0.001	-0.154	0.878	0.000	202
	change_ln_spot	-0.000	0.005	-0.034	0.973	0.000	202
	change_ois_2y	-0.003	0.005	-0.619	0.537	0.194	202
	change_ois_2y	0.245	0.043	5.708	0.000	0.194	202
	change_ois_5y	-0.012	0.007	-1.789	0.075	0.058	202
	change_ois_5y	0.151	0.051	2.964	0.003	0.058	202
	change_ois_10y	-0.014	0.007	-1.881	0.061	0.019	202
	change_ois_10y	0.089	0.050	1.795	0.074	0.019	202
	change_forward_2y	-0.000	0.001	-0.273	0.785	0.003	202
	change_forward_2y	0.004	0.005	0.993	0.322	0.003	202
	change_forward_5y	-0.001	0.001	-0.996	0.320	0.012	202
	change_forward_5y	0.009	0.005	1.790	0.075	0.012	202
	change_forward_10y	-0.140	0.075	-1.880	0.062	0.019	202
	change_forward_10y	0.891	0.497	1.794	0.074	0.019	202
gdp models	change_ln_spot	-0.001	0.002	-0.410	0.683	0.086	68
	change_ln_spot	-0.006	0.002	-3.320	0.001	0.086	68
	change_ois_2y	0.003	0.012	0.206	0.837	0.113	68
	change_ois_2y	0.067	0.022	2.998	0.004	0.113	68
	change_ois_5y	0.017	0.013	1.312	0.194	0.079	68
	change_ois_5y	0.043	0.022	1.958	0.054	0.079	68
	change_ois_10y	0.026	0.012	2.082	0.041	0.038	68
	change_ois_10y	0.032	0.024	1.357	0.179	0.038	68
	change_forward_2y	-0.001	0.002	-0.264	0.793	0.066	68
	change_forward_2y	-0.006	0.002	-2.708	0.009	0.066	68
	change_forward_5y	-0.001	0.002	-0.308	0.759	0.054	68
	change_forward_5y	-0.006	0.003	-2.231	0.029	0.054	68
	change_forward_10y	0.255	0.124	2.059	0.043	0.036	68
	change_forward_10y	0.314	0.237	1.323	0.190	0.036	68
south africa unemployment models	change_ln_spot	-0.001	0.002	-0.570	0.571	0.008	49
	change_ln_spot	0.001	0.000	1.495	0.142	0.008	49
	change_ois_2y	0.009	0.008	1.124	0.267	0.019	49
	change_ois_2y	0.004	0.002	2.268	0.028	0.019	49
	change_ois_5y	0.012	0.010	1.225	0.227	0.004	49
	change_ois_5y	0.002	0.002	1.144	0.258	0.004	49
	change_ois_10y	0.008	0.011	0.796	0.430	0.001	49
	change_ois_10y	0.002	0.003	0.567	0.574	0.001	49
	change_forward_2y	-0.001	0.002	-0.518	0.607	0.007	49
	change_forward_2y	0.001	0.000	1.448	0.154	0.007	49
	change_forward_5y	-0.001	0.002	-0.662	0.511	0.009	49
	change_forward_5y	0.001	0.000	1.429	0.160	0.009	49
	change_forward_10y	0.083	0.106	0.782	0.438	0.001	49
	change_forward_10y	0.016	0.027	0.581	0.564	0.001	49

Table A5: US monetary policy surprises models

surprise type	dependant variables	estimate	std.error	statistic	p.value	r.squared	nobs
target models	change_ln_spot	0.001	0.003	0.590	0.556	0.003	123
	change_ois_2y	-0.035	0.020	-1.798	0.075	0.028	123
	change_ois_5y	-0.029	0.019	-1.561	0.121	0.020	123
	change_ois_10y	-0.028	0.015	-1.798	0.075	0.020	123
	change_forward_2y	0.001	0.003	0.521	0.604	0.002	123
	change_forward_5y	0.001	0.002	0.413	0.680	0.001	123
	change_forward_10y	-0.277	0.156	-1.773	0.079	0.020	123
forward guidance models	change_ln_spot	0.004	0.001	2.591	0.011	0.067	123
	change_ois_2y	-0.052	0.007	-7.620	0.000	0.252	123
	change_ois_5y	-0.044	0.009	-4.787	0.000	0.176	123
	change_ois_10y	-0.030	0.009	-3.257	0.001	0.091	123
	change_forward_2y	0.003	0.001	2.302	0.023	0.046	123
	change_forward_5y	0.002	0.001	1.597	0.113	0.024	123
	change_forward_10y	-0.293	0.091	-3.227	0.002	0.089	123
central bank information models	change_ln_spot	-0.002	0.001	-1.555	0.123	0.020	123
	change_ois_2y	-0.009	0.015	-0.611	0.543	0.007	123
	change_ois_5y	-0.003	0.016	-0.164	0.870	0.001	123
	change_ois_10y	0.003	0.016	0.207	0.836	0.001	123
	change_forward_2y	-0.002	0.001	-1.339	0.183	0.016	123
	change_forward_5y	-0.001	0.001	-0.644	0.521	0.003	123
	change_forward_10y	0.030	0.156	0.192	0.848	0.001	123
lsap models	change_ln_spot	0.002	0.002	1.184	0.239	0.014	123
	change_ois_2y	-0.010	0.012	-0.806	0.422	0.005	123
	change_ois_5y	-0.045	0.012	-3.666	0.000	0.097	123
	change_ois_10y	-0.050	0.011	-4.581	0.000	0.133	123
	change_forward_2y	0.002	0.002	0.998	0.320	0.009	123
	change_forward_5y	0.001	0.002	0.708	0.480	0.004	123
	change_forward_10y	-0.495	0.108	-4.574	0.000	0.131	123